

Unit-1

Mathematical Logic

19/Aug/24

* Statement or Proposition.

A declarative sentence which is true or false but not both is called statement.

There are 2 types

1] Primary or Atomic Statement

A declarative sentence which cannot be broken down into simple sentences is called primary or atomic statement.

2] Compound statement or Molecular Statement

A statement which contains two or more primary sentences and connectives.

$(\neg, \vee, \rightarrow, \leftrightarrow, \vdash)$ are called compound or molecular statement.

* Connectives

The words or symbols which are used to make a compound statements are called connectives.

Ex: $\neg, \vee, \rightarrow, \leftrightarrow, \vdash$

Connectives are 5 types

Conjunction

$p \wedge q$ is $p \text{ and } q$ which is read as p and q
 $p \wedge q$ has truth value T only when both p and q
have truth value T otherwise it has truth
value F (it acts as and (EE) gate)

Truth Table

p	q	$p \wedge q$
-----	-----	--------------

T	T	T
-----	-----	-----

T	F	F
-----	-----	-----

F	T	F
-----	-----	-----

F	F	F
-----	-----	-----

2] Disjunction

The disjunction of two statements
 $p \vee q$ is $p \text{ or } q$ which is read as p or q
 $p \vee q$ has truth value F only when both
 p and q have truth values F otherwise
it has truth value T

Truth Table

p	q	$p \vee q$
-----	-----	------------

T	T	T
-----	-----	-----

T	F	T
-----	-----	-----

F	T	T
-----	-----	-----

F	F	F
-----	-----	-----

3] Conditional Statement

the Conditional statement of p, q is $p \rightarrow q$ which is read as Implies of (con) if p then q , (con) q whenever p .

$p \rightarrow q$ has truth value F only when p is T and q is F otherwise it has truth value T

Truth Table:-

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Bi-Conditional Statement

the Bi-Conditional statement of p, q is $p \leftrightarrow q$ which is read as p if and only if q

$p \leftrightarrow q$ has truth table value T when both p and q have same truth value otherwise it has truth value F

Truth Table

P	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Negation

A Negation of a statement is general formed by introducing the word NOT at a proper place in the given statement.

It is denoted by $\neg$ or $\sim$

Truth Table

P	$\neg p$
T	F
F	T

Construct truth Table for following statement $(p \vee q) \wedge (\neg p \vee \neg q)$

P	q	r	$(p \vee q)$	$\neg p$	$\neg r$	$(\neg p \vee \neg r)$	$(p \vee q) \wedge (\neg p \vee \neg r)$
T	T	T	T	F	F	F	F
T	T	F	T	F	T	T	T
T	F	T	T	F	F	F	F
T	F	F	T	F	T	T	T
F	T	T	T	T	F	T	T
F	T	F	T	T	T	T	T

F	F	T	f	T	F	T	F
F	f	F	F	T	T	T	F

$$[(p \vee q) \wedge (\neg r)] \leftrightarrow (q \rightarrow r)$$

P	Q	r	$(p \vee q)$	$(\neg r)$	$(p \vee q) \wedge (\neg r)$	$(q \rightarrow r)$	$[(p \vee q) \wedge (\neg r)] \leftrightarrow (q \rightarrow r)$
T	T	T	T	F	F	T	F
T	T	F	T	T	T	F	F
T	F	T	T	F	F	T	F
T	F	F	T	T	T	T	T
F	T	T	T	F	F	T	F
F	T	F	T	T	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	T	F	T	F

Inverse Converse and Contrapositive

the inverse of p implies q is $p \rightarrow q$ is $\neg p \rightarrow \neg q$

the converse of p implies q is $p \rightarrow q$ is $q \rightarrow p$

the contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$

problem: What are the Inverse Converse and Contrapositive of Implication "if Today is a holiday Then I will go for a movie"

Let p : Today is a holiday

q : I will go for a movie then $p \rightarrow q$

(i) Inverse is $\neg p \rightarrow \neg q$

that "if today is not a holiday Then I will not go for a movie"

(ii) Converse is $q \rightarrow p$

"if I will go for a movie Then Today is a holiday"

(iii) Contrapositive $\neg q \rightarrow \neg p$

"if I will not go for a movie Then Today is not a holiday"

② "The home team does not win whenever it is raining"

We know That "q whenever p" means " $p \rightarrow q$ "

Let p : It is raining

q : The home team does not win

(i) Inverse is $\neg p \rightarrow \neg q$

"if it is not raining then the home team wins!"

(ii) Converse is $q \rightarrow p$

"if the home team does not win then it is raining!"

(iii) Contrapositive is $\neg q \rightarrow \neg p$

"if the home team wins then it is not raining"

if ΔABC is right triangle then $(AB)^2 + (BC)^2 = (AC)^2$

(i) Inverse is $\neg p \rightarrow \neg q$

if ΔABC is not right angle then $(AB)^2 + (BC)^2 \neq (AC)^2$

(ii) Converse is $q \rightarrow p$

"if $(AB)^2 + (BC)^2 = (AC)^2$ then ΔABC is right triangle"

(iii) Contrapositive is $\neg q \rightarrow \neg p$

"if $(AB)^2 + (BC)^2 \neq (AC)^2$ then ΔABC is not right triangle"

Tautology (or) universally valid formula

A Compound statement which is always true for all possible Truth values of proposition that occur in it is called tautology proposition

Ex: $p \vee \neg p$ is a Tautology.

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Contradiction:

A Compound Statement which is always false for all possible Truth of proposition that occur in it is called Contradiction

Ex: $p \wedge \neg p$ is a Contradiction

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Contingency:

A Statement which is neither tautology nor Contradiction is called Contingency.

Q. Determine which of the following statements formulas or Tautology or Contradiction

1. $(p \rightarrow (p \vee q))$

2. $(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$

3. $\neg [p \vee (q \wedge r)] \leftrightarrow [\neg p \vee \neg q] \wedge [\neg p \vee \neg r]$

IV. A statement formula is well formed formula if and only if it can be obtained by finite application of above three rules

P Indicate which of the following formulas are well formed

① $\neg(p \wedge q) \rightarrow$ it is a well formed formula

② $(p \rightarrow \neg(p \vee q)) \rightarrow$ it is a well formed formula

③ $p \rightarrow q$ it is not well formed formula because it has no parenthesis

Equivalent (or) Logically Equivalent
The two compound component statements P and Q are said to be equivalent if they have same truth tables. It is denoted by $p \equiv q$ (or) $p \Leftrightarrow q$

Problem: Show that the proposition $p \rightarrow q$ and $\neg p \vee q$ are logically equivalent.

(or) Show that $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$

p	q	$p \rightarrow q$	$\neg p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Since $(p \rightarrow q)$ and $(\neg p \vee q)$ have same truth tables $p \rightarrow q \equiv (\neg p \vee q)$

Show That $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ is $2^3 = 8$

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$(p \vee q)$	$(p \vee r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	F	T	F
T	F	F	F	T	F	F	F
F	T	T	T	T	T	T	T
F	T	F	F	T	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Equivalent formulas

1. Commutative Laws:-

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

2. Associative Laws:

$$p \vee (q \vee r) \equiv (p \vee q) \vee r$$

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$$

3. Idempotent Laws:

$$p \vee p \equiv p$$

$$p \wedge p \equiv p$$

4. Negation Laws

$$p \vee \neg p \equiv T$$

$$p \wedge \neg p \equiv F$$

5. De Morgan Laws

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

6. Distributive Laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

7. Double Negation Law:

$$\neg(\neg p) \equiv p$$

8. Law of Implication

$$(p \rightarrow q) \equiv (\neg p \vee q)$$

9. Identity Laws :

$$p \wedge T \equiv p$$

$$p \vee F \equiv p$$

$$a \times 1 = a$$

$$a + 0 = a$$

10. Dominant Laws

$$p \vee T \equiv T$$

$$p \wedge F \equiv F$$

11. Absorption Law

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

Without using truth tables show that $\neg[p \vee (\neg p \wedge q)]$ and $(\neg p \wedge \neg q)$ are logically equivalent

$$\neg[p \vee (\neg p \wedge q)]$$

$$\equiv \neg p \wedge \neg(\neg p \wedge q) \text{ by De Morgan Law}$$

$$\equiv \neg p \wedge (p \vee \neg q) \text{ by double negation and De Morgan Law}$$

$$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \text{ by distributive Law}$$

$\equiv \neg(\neg p \wedge \neg q)$ by negation law

$\neg p \wedge \neg q$ by Identity law

Show that $(p \rightarrow q) \rightarrow q \iff p \vee q$

$\neg(p \rightarrow q) \vee q$ by law of implication

$\neg(\neg p \vee q) \vee q$ " Law of implication

$(p \wedge \neg q) \vee q$ by double negation and DeMorgan's laws

$(p \wedge \neg q) \vee q$

$(p \vee q) \wedge (\neg q \vee q)$ by distributive law

$(p \vee q) \wedge \neg$ by negation law $a \times 1 = a$

$p \vee q$ by Identity law

Show that $p \rightarrow (q \rightarrow r) \iff (p \wedge q) \rightarrow r$

$p \rightarrow (\neg q \vee r)$

$\neg p \vee (q \rightarrow r)$ by law of implication

$\neg p \vee (\neg q \vee r)$ " " " "

Show that $\neg(p \wedge q) \rightarrow [\neg p \vee (\neg p \vee q)] \Leftrightarrow (\neg p \vee q)$
without using truth tables

Given that $\neg(p \wedge q) \rightarrow [\neg p \vee (\neg p \vee q)] \Leftrightarrow (\neg p \vee q)$

$\neg(p \wedge q) \rightarrow (\neg p \vee (\neg p \vee q))$ by Law of
Implication

$\neg(\neg(p \wedge q)) \vee [\neg p \vee (\neg p \vee q)]$ by double negation
and Associative Laws

$(p \wedge q) \vee [\neg p \vee q]$ by Idempotent Law

$[p \vee (\neg p \vee q)] \wedge [q \vee (\neg p \vee q)]$ by distributive
law

$[(p \vee \neg p) \vee q] \wedge [q \vee (\neg p \vee p)]$ by Associative and
Commutative Law

$(\top \vee q) \wedge [(q \vee q) \vee \neg p]$ by negation Laws and
Associative Laws

$\top \wedge (q \vee \neg p)$ by dominant Law and Idempotent
Law

$q \vee \neg p$ by Identity Law

$\neg p \vee q$ by Commutative Law

Show that $(p \vee q) \wedge [\neg p \wedge (\neg p \wedge q)] \equiv (p \wedge q)$ without
using truth tables

$(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q))$

$(p \vee q) \wedge ((\neg p \wedge \neg p) \wedge q)$ by Associative Law

$(p \vee q) \wedge (\neg p \wedge q)$ by Idempotent Law

$p \wedge (\neg p \wedge q) \vee (q \wedge (\neg p \wedge q))$ by distribution Law

$(p \wedge (\neg p) \wedge q) \vee (q \wedge q) \wedge \neg p$

$F \wedge q \vee q \wedge \neg p$ by negation Law
and Idempotent Law

$$\neg p \wedge q$$

Duality Law and (or) Dual formulas:

2 formulas x and x^* are said to be duals of each other if one can be obtained from the other by replacing each $\wedge$ by $\vee$, each $\vee$ by $\wedge$, each $\neg$ by F and each F by T .

Write duals of following statements/formulas

① $(p \vee q) \wedge r$

② $(p \wedge q) \vee T$

③ $(p \vee q) \wedge (p \vee \neg(q \wedge \neg s)) \vee F$

~~Dual of $(p \wedge q) \vee r$~~ ① Dual of $(p \vee q) \wedge r$ is $(p \wedge q) \vee r$

~~Dual of $(p \vee q) \wedge T$~~ ② Dual of $(p \wedge q) \vee F$ is $(p \vee q) \wedge F$

③ Dual of $(p \vee q) \wedge (p \vee \neg(q \wedge \neg s))$ is $(p \wedge q) \vee (p \wedge \neg(q \vee \neg s)) \wedge T$

Tautological Implication

A statement P is said to be tautologically implies a statement Q if $P \rightarrow Q$ is Tautology
It is denoted by $P \Rightarrow Q$ ($\Rightarrow$ = Tautologically implies)

Show that $[P \rightarrow (P \rightarrow R)] \Rightarrow [(P \rightarrow Q) \rightarrow (P \rightarrow R)]$

P	Q	R	$P \rightarrow R$	$P \rightarrow (P \rightarrow R)$	$(P \rightarrow Q)$	$(P \rightarrow Q) \rightarrow (P \rightarrow R)$
T	T	T	T	T	T	T
T	T	F	F	F	T	F
T	F	T	T	T	F	T
T	F	F	F	F	F	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

Since all truth values are Truth
 $P \rightarrow (P \rightarrow R) \rightarrow [(P \rightarrow Q) \rightarrow (P \rightarrow R)]$ is Tautology

$\therefore [P \rightarrow (P \rightarrow R)] \Rightarrow [(P \rightarrow Q) \rightarrow (P \rightarrow R)]$ ✓

Show that $P \wedge (P \rightarrow Q) \Rightarrow Q$
 $P \wedge (P \rightarrow Q) \Rightarrow Q$ is Tautology

$P \wedge (P \rightarrow Q) \Rightarrow Q$ is Tautology

$P \wedge (P \rightarrow Q) \Rightarrow Q$ is Tautology

T	T
T	F
F	T
F	F

T	T
T	F
T	T
T	F
T	T
T	F
T	T
T	F

* Elementary

the sum of negations is

* Let p and

Q $P \vee Q$, $P \vee$
factor:

Any part of which is

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$p \wedge (p \rightarrow q) \Rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Since all truth values are Truth

$p \rightarrow q, p \rightarrow q \rightarrow$
 $(p \rightarrow q) \rightarrow p \rightarrow q$
 $p \wedge (p \rightarrow q) \rightarrow q$ is Tautology
 $p \wedge (p \rightarrow q) \Rightarrow q$

T
T
T
T
T
T
T

Elementary product

The product of statement variable and their negation is called Elementary product

* Let p, q be two statement variable

Then $p \wedge q, p \wedge \neg p, q \wedge \neg q, p \wedge \neg p \wedge \neg q$ - - -

are Elementary products

* Elementary sum

The sum of statement variable and their negations is called Elementary sum

* Let p and q be two statement variable then p and

$q, p \vee q, p \vee \neg p, q \vee \neg q, p \vee \neg p \vee \neg q$

factors:

Any part of an Elementary sum or product which is itself the Elementary sum or product

is called factor of Elementary Sum or product

Ex: $p \vee \neg p$, $p \vee \neg q$, $\neg p \vee \neg q$ are same factors of $p \vee \neg p \vee \neg q$

Disjunctive Normal form:- (Sum of product)
the sum of Elementary products which is Equivalent to given statement formula is called disjunctive normal form of given formula.

Conjunctive normal forms (Product of sum)
the product of Elementary sum which is Equivalent to given statement formula is called Conjunctive normal form of given formula

Obtain disjunctive normal form of (DNF) of $p \wedge (p \rightarrow q)$

$$p \wedge (p \rightarrow q)$$

$$\equiv p \wedge (\neg p \vee q) \text{ by Law of Implication}$$

$$\equiv (p \wedge \neg p) \vee (p \wedge q) \text{ by Distributive Law}$$

$\rightarrow$ it is required D.N.F.

Obtain Conjunctive normal form of (CNF) of $p \wedge (p \rightarrow q)$

$$p \wedge (p \rightarrow q)$$

$$\equiv p \wedge (\neg p \vee q) \text{ by Law of Implication}$$

$\hookrightarrow$ it is required CNF

Obtain disjunctive normal form of $p \rightarrow [(p \rightarrow q) \wedge \neg(\neg q \vee \neg p)]$

$$\equiv \neg p \vee [(p \rightarrow q) \wedge \neg(\neg q \vee \neg p)] \text{ by Law of Implication}$$

$$\equiv \neg p \vee [(p \vee \neg q) \wedge (q \wedge p)] \text{ by Law of Implication}$$

$$\equiv \neg p \vee [(\neg p \wedge (q \wedge p)) \vee (q \wedge (\neg p \wedge p))] \text{ by De Morgan's Law, Double negation, distributive Law}$$

$$\neg p \vee (\neg p \wedge q \wedge p) \vee (q \wedge \neg p \wedge p) \text{ which is required D.N.F.}$$

Obtain CNF of $p \rightarrow [(p \rightarrow q) \wedge \neg(\neg q \vee \neg p)]$

$$\neg p \vee [(p \rightarrow q) \wedge \neg(\neg q \vee \neg p)] \text{ by Law of Implication}$$

$$\equiv \neg p \vee [(p \vee \neg q) \wedge (q \wedge p)] \text{ by Law of Implication, De Morgan's Law, Double negation}$$

$$\equiv \neg p \vee [(q \vee p) \wedge q] \wedge p \text{ by Associative Law}$$

$$\neg p \vee [(p \vee q) \vee (q \wedge q)] \wedge p \text{ by distributive Law}$$

$$\neg p \vee [(p \vee q) \vee (q \wedge p)]$$

$$\equiv \neg p \vee [q \wedge p] \text{ by Absorption Laws}$$

$$\equiv (\neg p \vee q) \wedge (\neg p \vee p) \text{ by distributive Law}$$

Obtain DNF of and CNF of $\neg(p \vee q) \leftrightarrow (p \wedge q)$

$$[\neg(p \vee q) \rightarrow (p \wedge q)] \wedge [(p \wedge q) \rightarrow \neg(p \vee q)] \equiv p \leftrightarrow q \equiv p \rightarrow q \wedge (q \rightarrow p)$$

$$\neg(\neg p \vee \neg q) \vee (p \wedge q) \wedge [\neg(p \wedge q) \vee \neg(p \vee q)] \text{ by Law of Implication}$$

$$\begin{aligned}
&= [(p \vee q) \wedge (p \wedge q)] \wedge [(\neg p \vee \neg q) \vee (\neg p \wedge \neg q)] \text{ by De Morgan's laws} \\
&= [p \vee (q \wedge p \wedge q)] \wedge [\neg p \vee (\neg q \wedge \neg p \wedge \neg q)] \text{ by Associative laws} \\
&= (p \vee q) \wedge (\neg p \vee \neg q) \text{ by Absorption law} \\
&= [p \wedge (\neg p \vee \neg q)] \vee [q \wedge (\neg p \vee \neg q)] \text{ by distributive laws} \\
&= [(p \wedge \neg p) \vee (p \wedge \neg q)] \vee [(q \wedge \neg p) \vee (q \wedge \neg q)] \text{ " " } \\
&= (p \wedge \neg p) \vee (p \wedge \neg q) \vee (q \wedge \neg p) \vee (q \wedge \neg q) \text{ which is}
\end{aligned}$$

required D.N.F

ENF $\neg(p \vee q) \leftrightarrow (p \wedge q)$

$(p \vee p) \vee q$
 $(p \vee q) \vee p$

$(p \vee p) \vee p$
 $(p \vee q) \vee p$

$p \vee (q \vee p)$
 $(p \vee q) \vee p$

$$\begin{aligned}
&= [\neg(p \vee q) \rightarrow (p \wedge q)] \wedge [(p \wedge q) \rightarrow \neg(p \vee q)] \\
&= [(p \vee q) \vee (p \wedge q)] \wedge [\neg(p \wedge q) \vee \neg(p \vee q)] \text{ by Law of implication} \\
&= [(p \vee q) \vee (p \wedge q)] \wedge [\neg p \vee \neg q \vee (\neg p \wedge \neg q)] \text{ by De Morgan's law} \\
&= [(p \vee q) \vee p] \wedge [(p \vee q) \vee q] \wedge [(\neg p \vee \neg q) \vee \neg p] \wedge [(\neg p \vee \neg q) \vee \neg q] \text{ by}
\end{aligned}$$

which is required CNF

Min Terms:
 For a given number of variables the min term consist of conjunctions in which each variable or its negation appears only once but not both

* for Two Variables p and q The min terms are
 $p \wedge q, p \wedge \neg q, \neg p \wedge q, \neg p \wedge \neg q$

max terms:

for a given number of variables The max term consist of disjunction in which each variable or its negation appears only once but not both.

* for Two Variable p and q The max terms are
 $p \vee q, p \vee \neg q, \neg p \vee q, \neg p \vee \neg q$

Principle disjunctive normal form $\Sigma \Pi$

for a given formula, an Equivalent formula consisting of disjunction of min terms only is known as its Principle disjunctive normal form, sum of product

^{distib}
^{utiv law} Canonical form

Principle Conjunctive Normal form ΠF

for a given formula an Equivalent formula consisting of conjunction of max terms only is known as its Principle conjunctive normal form, the product of sum of Canonical form $\Pi \Sigma$

Obtain ΠF of $(\neg p \vee q) \wedge (p \vee \neg q) \wedge (p \wedge q)$
 $(\neg p \vee q) \wedge (p \vee \neg q) \wedge (p \wedge q)$

$$\begin{aligned}
 & \neg p \vee q \\
 & \equiv (\neg p \wedge T) \vee (q \wedge T) \text{ by Identity Laws} \\
 & \equiv [\neg p \wedge (q \wedge \neg q)] \vee [q \wedge (p \vee \neg p)] \text{ by negation Law} \\
 & \equiv [(\neg p \wedge q) \wedge (\neg p \wedge \neg q)] \vee [(q \wedge p) \vee (q \wedge \neg p)] \text{ by distributive Law} \\
 & \equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (p \wedge q) \vee (\neg p \wedge q) \text{ by Commutative Law} \\
 & \equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (p \wedge q) \text{ by Idempotent Laws} \\
 & \text{Which is required PCNF}
 \end{aligned}$$

Obtain PCNF of $\neg(p \vee q)$

$$\equiv (\neg p \wedge \neg q) \text{ by Demorgan's Laws}$$

$$\equiv [\neg p \wedge F] \vee [\neg q \vee F] \text{ by Identity Law}$$

$$\equiv \neg p \wedge (q \wedge \neg q) \vee (\neg q \vee (p \wedge \neg p)) \text{ by negation Laws}$$

$$\equiv (\neg p \wedge q) \wedge (\neg p \wedge \neg q) \wedge (\neg q \vee p) \wedge (\neg q \vee \neg p) \text{ by distributive Law}$$

$$\equiv (\neg p \vee q) \wedge (\neg p \wedge \neg q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q) \text{ by Commutative Law}$$

$$\equiv (\neg p \vee q) \wedge (\neg p \vee \neg q) \wedge (p \vee \neg q) \text{ by Idempotent Law}$$

Obtain PCNF of $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$

Given that $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$

$$\equiv [(p \wedge q) \wedge T] \vee [(\neg p \wedge r) \wedge T] \vee [(q \wedge r) \wedge T] \text{ by Identity Law}$$

$$[(p \wedge q) \wedge (q \vee \neg q)] \vee [(\neg p \wedge r) \wedge (p \vee \neg p)] \vee [(q \wedge r) \wedge (p \vee \neg p)]$$

$$\equiv [(p \wedge q) \wedge (q \vee \neg q)] \vee [(\neg p \wedge r) \wedge (q \vee \neg q)] \vee [(q \wedge r) \wedge (p \vee \neg p)] \text{ by negation Law}$$

$$\begin{aligned}
&= [(p \wedge q) \wedge r] \vee [(p \wedge q) \wedge \neg r] \vee [(p \wedge r) \wedge q] \vee [(p \wedge r) \wedge \neg q] \vee \\
&\quad [(q \wedge r) \wedge p] \vee [(q \wedge r) \wedge \neg p] \rightarrow \text{by distributive law} \\
&= (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge r \wedge q) \vee (p \wedge r \wedge \neg q) \vee \\
&\quad (q \wedge r \wedge p) \vee (q \wedge r \wedge \neg p) \rightarrow \text{by commutative laws} \\
&= (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r) \vee (p \wedge r \wedge \neg q) \vee \\
&\quad (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \\
&= (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r) \vee (p \wedge r \wedge \neg q) \vee (p \wedge q \wedge \neg r) \\
&\quad \rightarrow \text{by Idempotent Law} \\
&\quad \text{which is required proof}
\end{aligned}$$

Obtain of proof of negation of $p \vee (p \wedge \neg q \wedge r)$

$$\begin{aligned}
&= p \vee (p \wedge \neg q \wedge r) \\
&\equiv p \wedge \neg (p \wedge \neg q \wedge r) \text{ by Demorgans Law} \\
&\equiv p \wedge (p \vee q \vee \neg r) \text{ by " " \& Double negation laws} \\
&\equiv (p \wedge p) \vee (p \wedge q) \vee (p \wedge \neg r) \text{ by distributive Law} \\
&\quad p \vee (p \wedge q) \vee (p \wedge \neg r) \text{ by negation laws} \\
&\equiv (p \wedge q) \vee (p \wedge \neg r) \text{ by Identity Laws}
\end{aligned}$$

Consider $(p \wedge q) \vee (p \wedge \neg r)$

Identity

$$\begin{aligned}
&\equiv [(p \wedge q) \wedge T] \vee [(p \wedge \neg r) \wedge T] \text{ by Identity Laws} \\
&\equiv (p \wedge q) \wedge (T \vee \neg r) \vee (p \wedge \neg r) \wedge (q \vee \neg q) \text{ negation law} \\
&\quad [(p \wedge q) \wedge (T) \vee (p \wedge q \wedge \neg r)] \vee [(p \wedge \neg r \wedge q) \vee (p \wedge \neg r \wedge \neg q)] \\
&\quad \rightarrow \text{by distributive}
\end{aligned}$$

$$\begin{aligned}
 &\equiv (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r) \text{ by Commutative laws} \\
 &\equiv (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \text{ by } p \vee p \equiv p \\
 &\quad \text{which is required} \\
 &\quad p.p \text{ N.F}
 \end{aligned}$$

Obtain p.c.n.f. of $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$

$$\begin{aligned}
 &\equiv (p \vee r) \wedge (\neg q \rightarrow p) \wedge (p \rightarrow q) \text{ by Law of Implication and biconditional} \\
 &\equiv (p \vee r) \wedge (\neg \neg q \vee p) \wedge (\neg p \vee q) \text{ by Law of implication} \\
 &\equiv (p \vee r) \wedge (q \vee p) \wedge (\neg p \vee q) \\
 &\equiv [(p \vee r) \vee f] \wedge [(q \vee p) \vee p] \wedge [(p \vee q) \vee f] \\
 &\quad \text{by Identity Laws} \\
 &\equiv (p \vee r) \vee (q \wedge \neg q) \wedge [(q \vee p) \vee (\neg r \wedge r)] \wedge \\
 &\quad [(p \vee q) \vee (\neg r \wedge r)] \text{ by negation Law} \\
 &\equiv [(p \vee r) \vee q] \wedge (p \vee \neg r \vee q) \wedge [(q \vee p) \vee \neg r] \wedge (q \vee p \vee r) \\
 &\quad \wedge [(p \vee q) \vee \neg r] \wedge (p \vee q \vee r) \text{ by distributive Law} \\
 &\equiv \cancel{[(p \vee r \vee q) \wedge p \vee r \vee \neg q]} \wedge \cancel{[(q \vee p \vee \neg r) \wedge p]} \\
 &\equiv (p \vee r \vee q) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee \neg r) \wedge (p \vee \neg q \vee r) \\
 &\quad (p \vee r \vee q) \wedge (p \vee \neg q \vee r) \text{ by Commutative Law} \\
 &\equiv (p \vee r \vee q) \wedge (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r) \\
 &\quad \text{by Idempotent Law}
 \end{aligned}$$

Obtain the product of sum canonical form for the following formula

$$(p \wedge q \wedge r) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg q \wedge \neg r)$$

$$\equiv (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r) \text{ by Commutative Law}$$

$$\equiv (p \vee \neg p) \wedge (q \wedge r) \vee [\neg p \wedge \neg q \wedge \neg r] \text{ by Law of distribution}$$

$$\equiv (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r) \text{ by negation Law}$$

$$\equiv (q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r) \text{ by Identity Laws}$$

$$q \wedge r \vee (\neg p \wedge \neg q \wedge \neg r) \wedge (\neg \vee (\neg p \wedge \neg q \wedge \neg r))$$

$$\equiv [(q \vee \neg p) \wedge (q \vee \neg q) \wedge (r \vee \neg r)] \wedge [(\neg p \vee \neg p) \wedge (\neg q \vee \neg q) \wedge (\neg r \vee \neg r)] \text{ by distributive Law}$$

$$\equiv (q \vee \neg p) \wedge (r) \wedge (q \vee \neg r) \wedge (\neg p \vee \neg p) \wedge (\neg q \vee \neg q) \wedge (\neg r \vee \neg r) \text{ by distributive Law}$$

$$\equiv (q \vee \neg p) \wedge (q \vee \neg r) \wedge (\neg p \vee \neg p) \wedge (\neg q \vee \neg q) \text{ Identity Law}$$

$$\equiv q \wedge [(q \vee \neg p) \vee F] \wedge [(q \vee \neg r) \vee F] \wedge [(\neg p \vee \neg p) \vee F] \wedge [(\neg q \vee \neg q) \vee F]$$

$$[(q \vee \neg p) \vee (\neg p \wedge \neg r)] \wedge [(q \vee \neg r) \vee (p \wedge \neg p)] \wedge [(\neg p \vee \neg p) \vee (q \wedge \neg q)] \wedge [(\neg q \vee \neg q) \vee (p \wedge \neg p)] \text{ by Identity Laws}$$

$$[(\neg p \vee \neg q) \vee (p \wedge \neg p)] \text{ by}$$

$$\equiv [(q \vee \neg p \vee r) \wedge (q \vee \neg p \vee \neg r)] \wedge [(q \vee \neg r \vee p) \wedge (q \vee \neg r \vee \neg p)] \wedge$$

$$[(\neg p \vee \neg q \vee p) \wedge (\neg p \vee \neg q \vee \neg p)] \wedge [(\neg p \vee \neg q \vee p) \wedge (\neg p \vee \neg q \vee \neg p)]$$

$$\equiv [(p \vee q \vee r) \wedge (p \vee q \vee \neg r)] \wedge [(p \vee q \vee r) \wedge (\neg p \vee q \vee r)] \wedge$$

$$[(\neg p \vee q \vee r) \wedge (\neg p \vee \neg q \vee r)] \wedge [(p \vee \neg q \vee r) \wedge (\neg p \vee \neg q \vee r)]$$

$$\equiv (\neg p \wedge r)$$

$$\equiv (\neg p \vee q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r)$$

P.D.N.F by Truth tables

1. Construct truth table for given problem
2. Write min terms corresponding to Each truth value True
3. Disjunction of above all min terms is called P.D.N.F

P.C.N.F by Truth Tables

1. Construct truth table for given problem
2. Write max terms corresponding to Each truth value False
3. Conjunction of above all max terms is called P.C.N.F

P.C.N.F from P.D.N.F

1. find P.D.N.F of given problem's
2. find P.D.N.F of $\neg S$ (i.e. disjunction of remaining min terms)
3. $P.C.N.F = \neg(P.D.N.F \text{ of } \neg S)$

* p.d.n.f from p.c.n.f

1. find p.c.n.f of given problems

2. find p.c.n.f of $\neg$ s (i.e. conjunction of remaining maxterms)

3. p.d.n.f = $\neg$ (p.c.n.f of $\neg$ s)

Q] Obtain p.d.n.f from p.c.n.f for the following using Truth tables:

$$(i) (p \vee \neg q) \rightarrow (p \rightarrow \neg q) = S$$

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$	$p \rightarrow \neg q$	$(p \vee \neg q) \rightarrow (p \rightarrow \neg q)$
T	T	F	F	F	F	T
T	F	F	T	T	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

min terms corresponding to each T: $p \wedge q, p \wedge \neg q, \neg p \wedge q$

$\therefore$ p.d.n.f of S = $(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$

Now p.d.n.f of $\neg$ s = disjunction of remaining min terms ($\neg p \wedge \neg q$)

p.c.n.f of S = $\neg$ (p.d.n.f of $\neg$ s)

$$\neg (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$$

$\neg p \vee \neg q$ by Demorgans Law

ii) $p \vee (\neg p \rightarrow (q \vee (\neg q \rightarrow r)))$

p	q	r	$\neg p$	$\neg q$	$\neg q \rightarrow r$	$q \vee (\neg q \rightarrow r)$	$p \rightarrow (q \vee (\neg q \rightarrow r))$	$p \vee q$
T	T	T	F	F	T	T	T	T
T	T	F	F	F	T	T	T	T
T	F	T	F	T	T	T	T	T
T	F	F	F	T	F	F	F	T
F	T	T	T	F	T	T	T	T
F	T	F	T	F	T	T	T	T
F	F	T	T	T	T	T	T	T
F	F	F	T	T	F	F	F	F

min terms corresponding to each T

$p \wedge q \wedge r, p \wedge q \wedge \neg r, p \wedge \neg q \wedge r, p \wedge \neg q \wedge \neg r, \neg p \wedge q \wedge r, \neg p \wedge q \wedge \neg r,$
 $\neg p \wedge \neg q \wedge r$

P.D.N.F of S = $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee$
 $(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r)$

Now P.D.N.F of $\neg S$ = disjunction of remaining min terms

$(\neg p \wedge \neg q \wedge \neg r) = \neg p \wedge \neg q \wedge \neg r$

P.C.N.F of S = $\neg(\text{P.D.N.F of } \neg S)$

$\neg(\neg p \wedge \neg q \wedge \neg r)$

$p \vee q \vee r$

Validity using Truth Tables (1) Valid Inference

Let $H_1, H_2, H_3, \dots, H_n$ be premises and C be conclusion. Then we say that C is valid (1) C follow logically from given premises. if $(H_1 \wedge H_2 \wedge \dots \wedge H_n) \rightarrow C$ is Tautology.

Q) Determine whether the conclusion C follows logically from given premises.

$$H_1: \neg p \quad H_2: p \leftrightarrow q \quad C: \neg(p \wedge q)$$

p	q	$\neg p$	$p \leftrightarrow q$	$H_1 \wedge H_2$	$\neg(p \wedge q)$	$H_1 \wedge H_2 \rightarrow C$
T	T	F	T	F	F	T
T	F	F	F	F	T	T
F	T	T	F	F	T	T
F	F	T	T	T	T	T

$\therefore H_1 \wedge H_2 \rightarrow C$ is tautology

$\therefore C$ follows logically from given premises

Rules of Inference

Rule P: A premise (data) may be introduced at any point in the derivation.

Rule T: A formula may be introduced at any point in the derivation.

* Implication formulas:

① Simplification: $p \wedge q \Rightarrow p$

$p \wedge q \Rightarrow q$

② Addition: $p \Rightarrow p \vee q$
 $q \Rightarrow p \vee q$

③ Conjunction: $p, q \Rightarrow p \wedge q$

④ modus ponens: $p \Rightarrow q, p \Rightarrow q$

⑤ modus Tollens: $p \Rightarrow q, \neg q \Rightarrow \neg p$

⑥ Disjunction, syllogismal
 $p \vee q, \neg p \Rightarrow q$
 $\neg p$

④ Transitive (a) Hypothetical syllogism

$$p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$$

⑧ Contrapositive

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Q] using Rules of inferences show that $r \vee s$ follows logically from the premises $\neg v d$, $(c \vee d) \rightarrow \neg h$, $\neg h \rightarrow (a \wedge \neg b)$, $(a \wedge \neg b) \Leftrightarrow (r \vee s)$

Steps	Statements	Reasons
1	$\neg v d$	Rule p
2	$(c \vee d) \rightarrow \neg h$	Rule p
3	$\neg h$	Rule T from 1 & 2 $p \rightarrow q, p \Rightarrow q$
4	$\neg h \rightarrow (a \wedge \neg b)$	Rule p
5	$(a \wedge \neg b)$	Rule T from 3 & 4 $p \rightarrow q, p \Rightarrow q$
6	$(a \wedge \neg b) \rightarrow (r \vee s)$	Rule p
7	$(r \vee s)$	Rule T from 5, 6 $p \rightarrow q, p \Rightarrow q$

Show that $R \wedge (p \vee q)$ is a valid conclusion from the premises $p \vee q$, $Q \rightarrow R$, $p \rightarrow M$ and $\neg M$.

Step	Statements	Reasons
1	$p \rightarrow M$	Rule p
2	$\neg M$	Rule p
3	$\neg p$	Rule T from 1 & 2 $p \rightarrow q, \neg q \Rightarrow \neg p$
4	$p \vee q$	Rule p
5	Q	Rule T from 3 & 4 $p \vee q, \neg p \Rightarrow q$
6	$Q \rightarrow R$	Rule p
7	R	Rule T from 5 & 6 $p \rightarrow q, p \Rightarrow q$
8	$R \wedge (p \vee q)$	Rule T from 4 & 7 $p, q \Rightarrow p \wedge q$

Show that $S \vee R$ is tautologically implied with $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow s)$

Steps	Statement	Reasons
1	$p \vee q$	Rule p
2	$\neg p \rightarrow q$	Rule T, from 1 & Law of Implication
3	$q \rightarrow s$	Rule p
4	$\neg p \rightarrow s$	Rule T, from 2 & 3
5	$\neg s \rightarrow p$	$p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$ Rule T, from 4 & Contrapositive Rule p
6	$p \rightarrow r$	
7	$\neg s \Rightarrow r$	Rule T, from 5 & 6
8	$s \vee r$	$p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$ Rule T from 7 & Law of implication

Show That $S \vee r$ is tautologically implied with
 $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow s)$

Steps	Statement	Reasons
1.	$p \vee q$	Rule P
2.	$\neg p \rightarrow q$	Rule T from 1 & Law of implication
3.	$q \rightarrow s$	Rule P
4.	$\neg p \rightarrow s$	Rule T from 2 & 3 & $p \rightarrow q$ $q \rightarrow r \Rightarrow p \rightarrow r$
5.	$\neg s \rightarrow p$	Rule T from 4 and Contra Positive Rule P
6.	$p \rightarrow r$	
7.	$\neg s \rightarrow r$	Rule T from 5 & 6 & $p \rightarrow r, q \rightarrow r \Rightarrow p \vee q \rightarrow r$
8.	$S \vee r$	Rule T from 7 and Law of Implication

$\therefore$ The Conclusion $S \vee r$ is valid

Rule C.P (C) Rule of Conditional proof (C) Deduction Theorem

$\Rightarrow$ If we can derive S from R and given set promises then we can derive $R \rightarrow S$ from given set of promises alone

Note: Rule $\rightarrow$ is generally used if the Conclusion is of the form $R \rightarrow S$

Show that $\neg p \rightarrow s$ can be derived from the premises $p \rightarrow (q \rightarrow s)$, $\neg p \vee p$ and q .

Steps	Statement	Reasons
1.	$\neg$	Rule $\neg$ (additional Premise)
2.	$\neg p \vee p$	Rule $\vee$
3.	p	Rule $\vee$ from 1, 2 $\neg p \vee p$
4.	$p \rightarrow (q \rightarrow s)$	$\neg p \rightarrow \neg$ Rule $\rightarrow$
5.	$q \rightarrow s$	Rule $\rightarrow$ from 3, 4 $p \rightarrow (q \rightarrow s)$, $p \Rightarrow q \rightarrow s$
6.	q	Rule $\vee$
7.	s	Rule $\rightarrow$ from 5, 6 $q \rightarrow s$, $q \Rightarrow s$
8.	$\neg p \rightarrow s$	Rule $\rightarrow$

$\neg p \rightarrow s$ is valid Conclusion

Test The Validity of following arguments

1. Soniya is watching TV.

2. if Soniya is watching TV, then she is not studying.

3. If she is not studying Then her father will not buy her Scooty

Therefore Soniya's will not buy a Scooty

Let us indicate the given statements as follows

p : Soniya is watching T.V

q : Soniya is studying

r : Soniya's will buy a Scooty Then

given problem becomes $p, p \rightarrow \neg q, \neg q \Rightarrow \neg r$

Steps

Statement

Reason

1.

p

Rule p

2.

$p \rightarrow \neg q$

Rule p

3.

$\neg q$

Rule p from 1, 2 $p \rightarrow \neg q, p \Rightarrow \neg q$

4.

$\neg q \Rightarrow \neg r$

Rule p

5.

$\neg r$

Rule p from 3, 4
 $p \rightarrow \neg q, p \Rightarrow \neg q$

$\therefore$ the argument is valid

If there was a meeting then catching the bus was difficult if they arrived on time then catching the bus was not difficult they arrived on time. Therefore there was no

meeting. Show That The Statement Constitutes A Valid argument
Let us indicate

p : There was a meeting

q : Catching the bus was difficult

r : They arrived on time Then The problem

became $p \rightarrow q, r \rightarrow \neg q, r \Rightarrow \neg p$

Steps	Statement	Reason
1.	$r \rightarrow \neg q$	Rule p
2.	r	Rule p
3.	$\neg q$	Rule T, from 1, 2 & $p \rightarrow q, p \Rightarrow q$
4.	$p \rightarrow q$	Rule p
5.	$\neg p$	Rule T, from 3, 4 { $p \rightarrow q, \neg q \Rightarrow \neg p$ }

The Argument is Valid

Show That The following Statements constitute A Valid argument

1. If A works hard then Either B or C will enjoy
2. If B enjoys then A will not ~~hard~~ work hard
3. If D enjoys then C will not
4. Therefore, If A works hard then D will not enjoy

Let us indicate the statement as follows

p : A works hard

q : B will enjoy

r : C " "

S : D " " They given problem becomes

$$p \rightarrow q \vee r, q \rightarrow \neg p, S \rightarrow \neg r \Rightarrow p \rightarrow \neg S$$

Steps	Statement	Reasons
1.	p	Rule p (Additional premise)
2.	$p \rightarrow p$ $p \rightarrow q \vee r$	Rule p
3.	$\neg p \vee r$	Rule T, from 1 & 2 $p \rightarrow q, p \Rightarrow q$
4.	$q \rightarrow \neg p$	Rule p
5.	$p \rightarrow \neg q$	Rule T, from 4 & Contrapositive
6.	$\neg q$	Rule T, from 1, 5 & $p \rightarrow q, p \Rightarrow q$
7.	$\neg p \vee r$	Rule $\vee$ from 3 & 6 & $p \vee q, \neg p \Rightarrow q$
8.	$S \rightarrow \neg r$	Rule p
9.	$\neg S$	Rule T from 7 & 8 & $p \rightarrow q, \neg q \Rightarrow \neg p$
10.	$p \rightarrow \neg S$	Rule cp

Consistency of premises:

Inconsistent: The set of premises $H_1, H_2, \dots, H_n$ is set to be inconsistent if $(H_1 \wedge H_2 \wedge \dots \wedge H_n)$ is called Contradiction

Consistent: The set of premises $H_1, H_2, \dots, H_n$ is set to be consistent if $(H_1 \wedge H_2 \wedge \dots \wedge H_n)$ is Tautology

Show that the premises $a \rightarrow (b \rightarrow c), d \rightarrow (b \wedge \neg c)$ and $a \wedge d$ are inconsistent

Steps	Statement	Reason
1	$a \wedge d$	Rule p
2	a	Rule T, from 1 & $p \wedge q \Rightarrow p$
3	d	Rule T, from 1 & $p \wedge q \Rightarrow q$
4	$a \rightarrow (b \rightarrow c)$	Rule p
5	$b \rightarrow c$	Rule T, from 2, 4 & $p \rightarrow q, p \Rightarrow q$
6	$d \rightarrow (b \wedge \neg c)$	Rule p
7	$b \wedge \neg c$	Rule T, from 3, 6 & $p \rightarrow q, p \Rightarrow q$
8	$(\neg(b \vee c)) \wedge (b \vee c)$	De Morgan's Law

9. $\neg(b \wedge \neg c)$ Rule 1, from 8 & De Morgan's Law

10. $(b \wedge \neg c) \wedge \neg(b \wedge \neg c)$ Rule 1 from 9 & $P \vee \neg P$

11. $\perp$ Rule 1 from 10 & negation Law

$\therefore$ given set of premisses is inconsistent

Show That The Premises $p \rightarrow q$, $p \rightarrow \neg q$, $q \rightarrow \neg r$, p are inconsistent

Steps

Statement

Reasons

1

$p \rightarrow q$

Rule P

2

p

Rule P

3

q

Rule T from 1, 2 &

4

$q \rightarrow \neg r$

$p \rightarrow q$, $p \Rightarrow q$
Rule P

5

$\neg r$

Rule T from 3, 4
& $p \rightarrow q$, $p \Rightarrow q$

6

$p \rightarrow \neg r$

Rule P

7

$(\neg p) \vee \neg r$

Rule 1, from 5, 6 &
 $p \rightarrow q$, $\neg q \Rightarrow \neg p$

8

$p \wedge \neg p$

Rule T from 2, 7
& $P \vee \neg P \Rightarrow P \wedge \neg P$

9

F

Rule T from 8
& negation Law

∴ given set of premises is inconsistent

Show that the following set of premises is inconsistent

1. if the Contract is Valid then John is Liable for penalty
2. if John Liable for penalty he will go bankrupt
3. if The bank will loan him money, he will not go bankrupt
4. As a matter of fact The Contract is Valid and The bank will loan him money

Let us indicate given statements as follows

p: The Contract is Valid

not $\rightarrow$ $\neg$

q: John is Liable for penalty

r: He will ^{not} go bankrupt

s: The bank will loan him money

Then given problem becomes

$p \rightarrow q, q \rightarrow r, s \rightarrow \neg r, p \wedge s$

Steps

Statement

Reason

1

$p \wedge s$

Rule p

2

p

Rule 1 from 1 $p \wedge q \rightarrow p$

3

s

Rule 1 from 1 $p \wedge q \rightarrow q$

4

$p \rightarrow q$

Rule p

5. q Rule T from 3, 4 & $p \rightarrow q$, $p \Rightarrow q$
6. $q \rightarrow r$ Rule p
7. r Rule T from 5, 6 & $p \rightarrow q$, $p \Rightarrow q$
8. $S \rightarrow \neg r$ Rule p
9. $\neg S$ Rule T from 7, 8 & $\neg \rightarrow q$, $\neg q \Rightarrow \neg p$
10. $S \wedge \neg S$ Rule T from 3, 9 & $p, q \Rightarrow p \wedge q$
11. F Rule T from 10 & negation law

given set of premises is inconsistent

Indirect method of proof

1. Take $\neg c$ as an additional premise
2. prove set of all premises is inconsistent
3. Hence, our assumption is wrong
4. $\therefore$ The conclusion is valid

problem:

Using indirect method so that $p \rightarrow q$, $q \rightarrow r$, $\neg(p \wedge r)$,

$$p \vee r \Leftrightarrow r$$

Indirect method: Take $\neg r$ as an additional premise and prove set of premises is inconsistent

Steps	Statement	Reasons
1.	$\neg r$	Rule p additional premise
2.	$q \rightarrow r$	Rule p
3.	$\neg q$	Rule $\neg$ from 1, 2 & $p \rightarrow q$ $\neg q \Rightarrow \neg p$
4.	$p \rightarrow q$	Rule p
5.	$\neg p$	Rule $\neg$ from 3, 4 & $p \rightarrow q$ $\neg q \Rightarrow \neg p$
6.	$p \vee r$	Rule p
7.	r	Rule $\vee$ from 5, 6 & $p \vee r$ $\neg p \Rightarrow r$
8.	$r \wedge \neg r$	Rule $\wedge$ from 1, 7 & $p, q \Rightarrow p \wedge q$
9.	F	Rule $\neg$ from 8 & negation law
10.	$\neg(p \wedge r)$	Rule p
11.	$F \wedge \neg(p \wedge r)$	Rule $\wedge$ from 9, 10 & $p, q \Rightarrow p \wedge q$
12.	F	Rule $\neg$ from 11 & $p \wedge F \Rightarrow F$

$\therefore$ our assumption is wrong
Hence, conclusion r is valid

using indirect method Show That $p \Rightarrow (q \vee r), q \rightarrow \neg p$
 $S \rightarrow \neg r, p \Rightarrow p \rightarrow \neg S$

Indirect method: Take $\neg(p \rightarrow \neg S) \Rightarrow \neg(\neg p \vee \neg S) = p \wedge S$ as an additional premise and prove set of premises is inconsistent

Steps	Statement	Reason
1.	$p \wedge s$	Rule p (additional premise)
2.	s	Rule T, from 1 & $p \wedge q \Rightarrow q$
3.	$s \rightarrow \neg r$	Rule p
4.	$\neg r$	Rule T from 2, 3 & $p \rightarrow q, p \Rightarrow q$
5.	$p \rightarrow (q \vee r)$	Rule p
6.	$q \vee p$	Rule p
7.	$q \vee r$	Rule T, from 5, 6 & $p \rightarrow q, p \Rightarrow q$
8.	$\neg r \vee q$	Rule T, from 7 & Commutative Law
9.	q	Rule T from 4, 8 & $p \vee q, \neg p \Rightarrow q$
10.	$q \rightarrow \neg p$	Rule p
11.	$\neg p$	Rule T from 9, 10 & $p \rightarrow q, p \Rightarrow q$
12.	$p \wedge \neg p$	Rule T from 6, 11 & $p \wedge q \Rightarrow p \wedge q$
13.	F	Rule T from 12 & negation law

our assumption is wrong Hence, conclusion $(p \rightarrow \neg s)$ is valid

2. Predicate: "x is a rational number" Here the first part x is called subject and second part is a "Rational number"

which refers to property that the subject than have is called predicate

* predicates are denoted Capital letters.

* subject are denoted by small letters

* The symbolic form of above statement of $R(x)$

Quantifiers:

Certain declarative sentence involve the words which indicate quantity such as all, some, one, none. These words are called quantifiers.

* Two Types

1. Universal Quantifier

2. Existential Quantifier

Universal Quantifier:

The quantifier 'all' is called universal quantifier

⇒ it is denoted by $(\forall x)$ and read as "for all x " or "for Every x " or "for Each x "

Existential Quantifier:

The quantifier 'some' is called Existential quantifier

⇒ it is denoted by $(\exists x)$ and read as "There exists x " or "for some x " or "There is at least one x "

Rule US (or) universal specification:
 $\frac{(\forall x) p(x)}{p(c)}$ where c is an arbitrary object

$(\forall x) p(x)$

$\therefore p(c)$ where c is an arbitrary object

Rule ES (or) Existential specification:
 $(\exists x) p(x)$

$(\exists x) p(x)$

$\therefore p(c)$ where c is an arbitrary object

Rule UG (or) universal generalization:-
 $\frac{p(c)}{\therefore (\forall x) p(x)}$

$p(c)$

$\therefore (\forall x) p(x)$

Rule EG (or) Existential generalization:-
 $\frac{p(c)}{\therefore (\exists x) p(x)}$

$p(c)$

$\therefore (\exists x) p(x)$

Note:- ① $\neg[(\forall x) A(x)] \equiv (\exists x) \neg A(x)$

② $\neg[(\exists x) A(x)] \equiv (\forall x) \neg A(x)$

Verify the validity of following arguments

"All men are mortal."

Socrates is a man

Therefore Socrates is mortal

$A(x)$: x is a man

$B(x)$: x is mortal

S : Socrates then the

Symbolic form is $(\forall x) [A(x) \rightarrow B(x)]$

$$\frac{A(s)}{\therefore B(s)}$$

Steps	Statement	Reason
1	$(\forall x) [A(x) \rightarrow B(x)]$	Rule p
2	$A(s) \rightarrow B(s)$	Rule 1, from 1 & Rule us
3	$A(s)$	Rule p
4	$B(s)$	Rule 1, from 2, 3 & $p \rightarrow q, p \Rightarrow q$

The Argument is valid

Establish The Validity of following arguments

All integers are rational numbers

Some " " and power of 2

Therefore some rational numbers ^{and} are powers of 2.

$A(x)$: x is an integer

$B(x)$: x is a rational number

$C(x)$: x is power of 2 then the

Symbolic form is $(\forall x) [A(x) \rightarrow B(x)]$

$$\neg (\exists x) [A(x) \wedge \neg B(x)]$$

$$\neg (\exists x) [B(x) \wedge \neg C(x)]$$

Steps	Statement	Reason
1	$(\forall x)[A(x) \rightarrow B(x)]$	Rule p
2.	$A(s) \rightarrow B(s)$	Rule T, from 1 & Rule US
3.	$(\exists x)[A(x) \wedge \neg C(x)]$	Rule p
4.	$A(s) \wedge \neg C(s)$	Rule T from 3 & Rule ES
5.	$A(s)$	Rule T from 4 & $p \wedge q \Rightarrow p$
6.	$\neg C(s)$	Rule T from 4 & $p \wedge q \Rightarrow q$
7.	$B(s)$	Rule T from 2 & $p \rightarrow q, p \Rightarrow q$
8.	$B(s) \wedge C(s)$	Rule T from 6, 7 & $p, q \Rightarrow p \wedge q$
9.	$(\exists x)[B(x) \wedge C(x)]$	Rule T from 8 & Rule EG

Verify the Validity following argument

Every living thing is a plant or an animal

Joe's gold fish is alive and it is not a plant.

All animals have hearts.

Therefore Joe's gold fish is ~~alive and it is not a~~ plant has a heart.

$A(x)$: x is living thing.

$B(x)$: x is a plant

$C(x)$: x is an animal

j : Joe's gold fish

$D(x)$: x has heart

Symbolic form is

$$(\forall x) [A(x) \rightarrow B(x) \vee C(x)]$$

$$A(j) \wedge \neg B(j)$$

$$(\forall x) [C(x) \rightarrow D(x)]$$

$$\therefore D(j)$$

Steps	Statement	Reason
1	$(\forall x) [A(x) \rightarrow B(x) \vee C(x)]$	Rule p
2	$\neg A(j) \rightarrow B(j) \vee C(j)$	Rule $\neg$, from 1 & Rule u.s.
3	$A(j) \wedge \neg B(j)$	Rule p
4	$A(j)$	Rule T, from 3 & $p \wedge q \Rightarrow p$
5	$\neg B(j)$	Rule T, from 3 & $p \wedge q \Rightarrow q$
6	$B(j) \vee C(j)$	Rule T, from 2, 4 & $p \rightarrow q, p \Rightarrow q$
7	$C(j)$	Rule T, from 5, 6 & $p \vee q, \neg p \Rightarrow q$
8	$(\forall x) [C(x) \rightarrow D(x)]$	Rule p
9	$C(j) \rightarrow D(j)$	Rule $\neg$, from 8 & Rule u.s.
10	$D(j)$	Rule T, $\neg, q, p \rightarrow q, p \Rightarrow q$

Show That $(\exists x) [p(x) \wedge Q(x)] \Rightarrow (\exists x) p(x) \wedge (\exists x) Q(x)$

Steps	Statement	Reason
1.	$(\exists x) [p(x) \wedge Q(x)]$	Rule P
2.	$p(c) \wedge Q(c)$	Rule T, from 1 & Rule ES
3.	$p(c)$	Rule T from 2 & $p \wedge q \Rightarrow p$
4.	$Q(c)$	Rule T from 2 & $p \wedge q \Rightarrow q$
5.	$(\exists x) p(x)$	Rule $\exists$, from 3 & Rule EG
6.	$(\exists x) Q(x)$	" " 4 " "
7.	$(\exists x) p(x) \wedge (\exists x) Q(x)$	Rule T, from 5, 6 & $p \wedge q \Rightarrow p \wedge q$